University of Cape Town
Department of Computer Science

Computer Science CSC3003S, 2024

Supplementary Exam

Paper 1 – Advanced Software Development

Answer ALL questions in both sections.

Each section MUST be written in **separate books**. Given your two books, please **write ASD1 at the top of the book** you will use for Section A: ASD1 and **write ASD2 at the top of the book** you will use for Section B: ASD2. If you use additional books, please label them also and place these inside the first book for the relevant section.

Marks: 50 *Marks per question are shown in brackets.*

Time: 90 minutes + 10 minutes reading time

The use of calculators is permitted.

Read through the entire exam paper before you start writing so that you can answer the easy questions first. The relative importance of the questions is indicated by the marks that are allocated to them. Reading time of **10 minutes** is included in the allocated time above.

Section A. ASD 1 – Project Management [25]

Please write this section in its own book. Write ASD1 at the top of the cover page of the book.

Question 1. [5]

Answer the following multiple-choice questions by writing down the NUMBER of the question and the LETTER of the best answer.

1. You need to test the functionality of the username and password authentication method in your program. What type of test do you use? [1]
   a) Integration
   b) Unit
   c) Performance
   d) All of the above
   e) None of the above
2. You are developing a mobile application that requires user authentication. Your client has asked for a prototype that demonstrates exactly how only the login process would function. What type of prototype will you use? [1]
   a) Horizontal, throw-away, paper prototype
   b) Horizontal, iterative/evolutionary, high fidelity interactive mock-up
   c) Vertical, throw-away; paper prototype
   d) Vertical, iterative/evolutionary, high fidelity interactive mock-up
3. In Scrum, which one of the following is the responsibility of the **Product Owner**? [1]
   a) Select tasks to be completed in the sprint.
   b) Prioritize the product backlog.
   c) Ensure that everyone follows the scrum rules and organization practices.
   d) Assigning tasks to individual team members.
4. Which of the following requirements is **not** SMART? [1]
   a) Uptime should be at least 99%.
   b) The application should be no more than 50MB in installed size.
   c) System response time should be less than one second.
   d) The system should be scalable.
5. Although your client did not request this, you decide to support face-recognition as part of the authentication process. What has happened? [1]
   a) Yak shaving
   b) Gold-plating
   c) Software entropy
   d) Refactoring

Question 2. [12]

Your company was hired to build a mobile app to support grocery deliveries for a large grocery chain. Their marketing team had already developed mock-ups of how they expect the application to look and you have signed a contract that clearly lays out a fixed timeline and the functional and non-functional requirements. They hired you because of your prior experience with e-commerce and delivery platforms, and you are expected to deliver a working product in four months. You will maintain the software for a fixed fee for two years after delivery of the software.

a) Give one reason why waterfall would be an appropriate approach for this project. [1]

b) You decide to rather take an agile approach, how will you decide what requirements to prioritize? [1]

c) Explain how the value of having a working product at all times reduces risk. [1]

d) Test driven development can be described as test-first development plus refactoring. Define test-first development and refactoring. Explain why you should use a TDD approach for the development of this software. [4]

e) You are considering using open-source libraries to support your implementation of the system. What are some of the issues (at least three) you need to consider and address before you finalise your decision? [3]

f) You realize that the problem you are solving is common and are considering making your code open-source. Name two tools you will use to support interaction with users and developers and explain how they will be used. [2]

Question 3. [8]

Your team is developing a system to help track students in a primary school aftercare program. In compliance with government regulations, each aftercare student must sign in when school ends, and their parent or guardian must sign them out when they are picked up. While most regular aftercare attendees are registered at the start of the year and pay a lump sum per term, the program also allows drop-ins who can be added to the system on demand, at the rate of R100 per day. The current system is paper-based, using a binder for each grade and a register for each week; parents find it tedious to find the right page to sign out, and this can be difficult if many parents arrive at once. The aftercare staff would like to also integrate the registers with a billing and payment processing system.

a) Develop one *user story* using the classic form (role, goal, reason) to capture the needs of the aftercare staff. [3]

b) Name key stakeholders in the development of this system. [2]

c) One of the key risks for this project is “developing the wrong user interfaces.” Explain why this a key risk for the project and give a 3M strategy for addressing the risk. [3]

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Section B. ASD 2 – Architecture and Code Quality [25]

Please write this section in its own book. Write ASD2 at the top of the cover page of the book.

Question 4. [25]

a) In a team project, we must ensure our part of the software being developed is reusable and maintainable/modifiable. Name any 1 other *non-run-time* property that is important when building good quality software. [1]

b) To be dependable, all processes in all stages of the software lifecycle must be *robust* (able to recover from failure). Name any one other property of dependable processes. [1]

c) Many tools can analyse code to help us improve code quality. What is code churn analysis? [1]

d) Which of the SOLID properties tells us to code in such a way that new functionality can be added without changing existing code? Answer DIP/ISP/LSP/OCP/SRP and give a brief ***reason*** for your choice. [1]

e) Which of the following sentences captures the Dependency Inversion Principle? Give the sentence number.

1. High-level modules, which provide complex logic, should be unaffected by changes in low-level modules.
2. Low-level modules should be unaffected by changes in high-level modules which have more complex logic. [1]

f) Jo designed an interface IProduct for her system. She is told today that some products do not have a size. IProduct has methods to set and return the size of a product. What change should she make to avoid violating ISP (the Interface Segregation Principle) ? [1]

g) UCT's HR department keeps track of many different kinds of employee: academics, gardeners, administrators, student tutors, etc. The details kept for each depends on the employee type i.e. the post/job they have.

1. Which OOP design pattern would you recommend using to record data when new staff are appointed? [1]
2. Suppose the statement below is run first to find out what type of staff the new member is:

   ```
   String empKind = JOptionPane.showInputDialog(null, "Type of post appointed to:");
   ```

   Give an example of the 2 Java statements that would follow this, in order to get that employee's details, when using your chosen design pattern.

   Choose any (meaningful) names for the variables and classes in your answer. Include a brief comment with each statement. [2]

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h) Consider the code below that uses the Composite pattern, and answer the questions that follow:

```
public class App {
    public static void main(String[ ] args) {
        Leaf leafA = new Leaf ("A");
        Leaf leafB = new Leaf("B");
        Composite compo = new Composite("X");
        compo.addPart(leafB);
        compo.addPart(leafA);
        compo.show();
    }
}
public class Leaf implements IPart {
    String name;
    public Leaf(String aName) {
        this.name = aName;
    }
    @Override
    public void show() {
        System.out.println(name);
    }
}
```

Give Java code for the Composite class, given that the Composite design pattern is being used here. Keep its instance variables and methods to the minimum required. [4]

i) RUP's 4+1 view of system architecture comprises use case scenarios and 4 views: the process view, the physical/deployment view and which 2 other views? Name *or* briefly describe each. [2]

j) Consider the diagram below. Give the number of the statement that is true. [1]

1. A is providing an interface that B is using.
2. B is providing an interface that A is using.



k) What is the difference between data processing applications and transaction processing applications? [1]

l) ABC sells its products to 5 companies located in England, France, Germany, Poland and Spain. Using a service oriented architecture, ABC is making its online site available in the 5 different languages, so clients can browse products and place orders in their own language. Outline any 2 services that ABC should provide, by briefly giving the function, input and output of each. [6]

m) "Whether SOA (service oriented architectures) or microservices is best, depends on what kind of system is being built". What kind of system is each most suited to? [2]

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